

Glossary of Key Terms – Interfaces

Voice Interface

A type of interface where users interact with a system by speaking.

Example: Asking a smart speaker to play music.

Speech Recognition

Technology that allows a computer to recognize and understand spoken words.

Example: A phone converting your speech into text messages.

Voice Assistant

A software assistant that responds to spoken commands or questions.

Example: Amazon Alexa answering questions about the weather.

Directed Dialogue

A conversation style where the system controls the interaction by asking specific questions.

Example: A flight booking system asking:
“What city do you want to fly to?”

Barge-in Feature

A feature that allows users to interrupt a voice system before it finishes speaking.

Example: Pressing or saying your choice before all menu options are read.

Guided Prompt

A message that helps users continue the interaction when the system does not understand them.

Example:
“Sorry, I didn’t understand. Did you mean Monday?”

Pen-based Device

A device that allows users to write or draw using a stylus or digital pen.

Example: Writing notes on a tablet using a stylus.

Smartpen

A digital pen that records what a user writes or draws.

Example: Livescribe Echo 2 Smartpen saving handwritten notes digitally.

Touchscreen

A display screen that responds to touch input.

Example: Using fingers to tap icons on a smartphone.

Multi-touch Surface

A touchscreen that can detect more than one touch at the same time.

Example: Pinching two fingers on a phone screen to zoom in.

Gesture

A movement of the hand, arm, or body used to communicate or control a system.

Example: Waving a hand to change slides during a presentation.

Touchless Interface

An interface that allows users to interact without physically touching a device.

Example: Using hand movements to control medical images in an operating room.

Computer Vision

Technology that allows computers to recognize and interpret visual information from cameras.

Example: A system detecting hand gestures through a webcam.

Microsoft Kinect

A motion-sensing technology that tracks body movement and gestures.

Example: Surgeons using gestures to move MRI images during surgery.

Haptic Interface

An interface that provides feedback through touch or vibration.

Example: A smartphone vibrating when receiving a notification.

Vibrotactile Feedback

Feedback given through vibrations.

Example: A smartwatch vibrating to alert the user.

Actuator

A component that creates movement or vibration in a device.

Example: The small motor inside a phone that makes it vibrate.

Multimodal Interface

An interface that combines multiple forms of interaction.

Example: Using speech and touch together on a smartphone.

Eye Gaze

Tracking where a person is looking.

Example: A computer detecting which part of the screen a user is focusing on.

Avatar

A digital representation of a user.

Example: A game character that copies the player's movements.

Shareable Interface

An interface designed for several people to use together.

Example: An interactive tabletop used during group discussions.

SmartBoard

A large interactive display that allows users to write or interact directly on the screen.

Example: Teachers writing notes on a digital classroom board.

Interactive Tabletop

A table-shaped digital display that supports group interaction.

Example: Several students moving digital objects on a shared table screen.

Augmented Reality (AR)

Technology that adds digital information or objects to the real world.

Example: Pokémon GO placing virtual characters in real environments.

Digital Augmentation

Digital content added to the real environment.

Example: Navigation arrows appearing on a car windshield.

Head-mounted Display

A wearable device that shows digital images in front of the user's eyes.

Example: AR headsets used for training or gaming.

Immersive Experience

An experience that makes users feel deeply involved in a virtual or digital environment.

Example: Wearing AR glasses that blend digital objects with the real world.